

# SMART INDIA HACKATHON 2026



## SAHARA

### Predictive Personnel Stress & Welfare Intelligence

- **Problem Statement ID – SIH26186 (Ministry of Home Affairs)**
- **Problem Statement Title - AI-Based Predictive Personnel Stress and Welfare Monitoring System for Uniformed Forces**
- **Theme - MedTech / BioTech / HealthTech**
- **PS Category - Software**
- **Team ID - 144977**
- **Team Name (Registered on portal) - TribeCoders**



# SAHARA: Privacy-First Predictive Welfare Intelligence

## ❖ Proposed Solution (Describe your Idea/Solution/Prototype)

**730** CAPF, NSG & AR personnel died by suicide and **55,555** left service in 2020-24 (MHA, Rajya Sabha Q.1036). **SAHARA** predicts **stress, burnout, emotional fatigue** and **welfare concerns** early from HRMS + voluntary app data, and turns every alert into human-led support. **Welfare, never surveillance or discipline.**



### Detailed explanation of the proposed solution

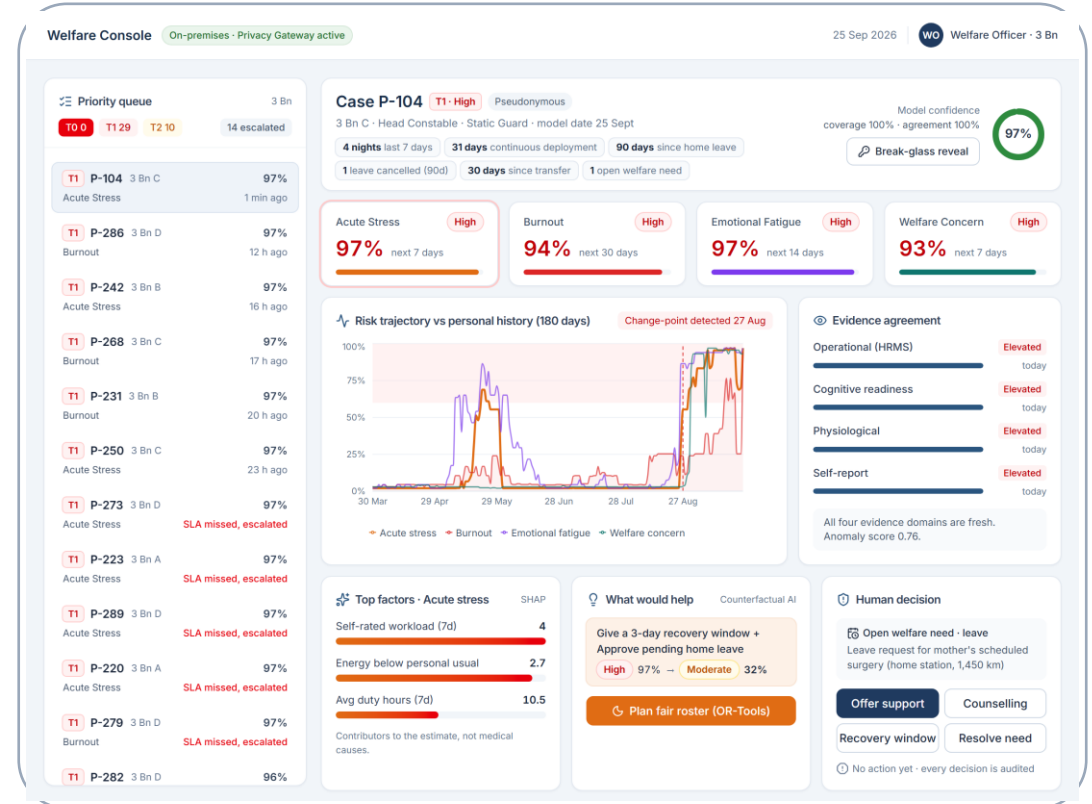
- **Mobile app:** consent per data type, 10-sec check-in, WHO-5, reaction test, camera heart-rate (on-device); Hindi, offline
- **HRMS analytics:** 26 features from leave, deployment, duty and night shifts, transfers, training, workload
- **AI engine:** personal baseline + anomaly + 4 calibrated XGBoost risk models + SHAP
- **Action:** tiered alerts, counterfactual advice, OR-Tools roster optimizer, recovery tracking

### How it addresses the problem

- **Late detection** → flagged early against each person's own baseline
- **Stigma & fear** → voluntary, pseudonymous; commanders see aggregates only
- **Alerts without action** → ranked interventions + fair roster + follow-up
- **Crisis moments** → Safe-Route to counsellor + Tele-MANAS 14416

### Innovation and uniqueness of the solution

- **Adaptive evidence:** one optional check lifts confidence 40% → 97% (live prototype)
- **'What would help' AI:** recovery window + leave → High to Moderate
- **My Data Mirror** + differential-privacy commander view
- **Purpose lock:** welfare data blocked from disciplinary use (403, audited)



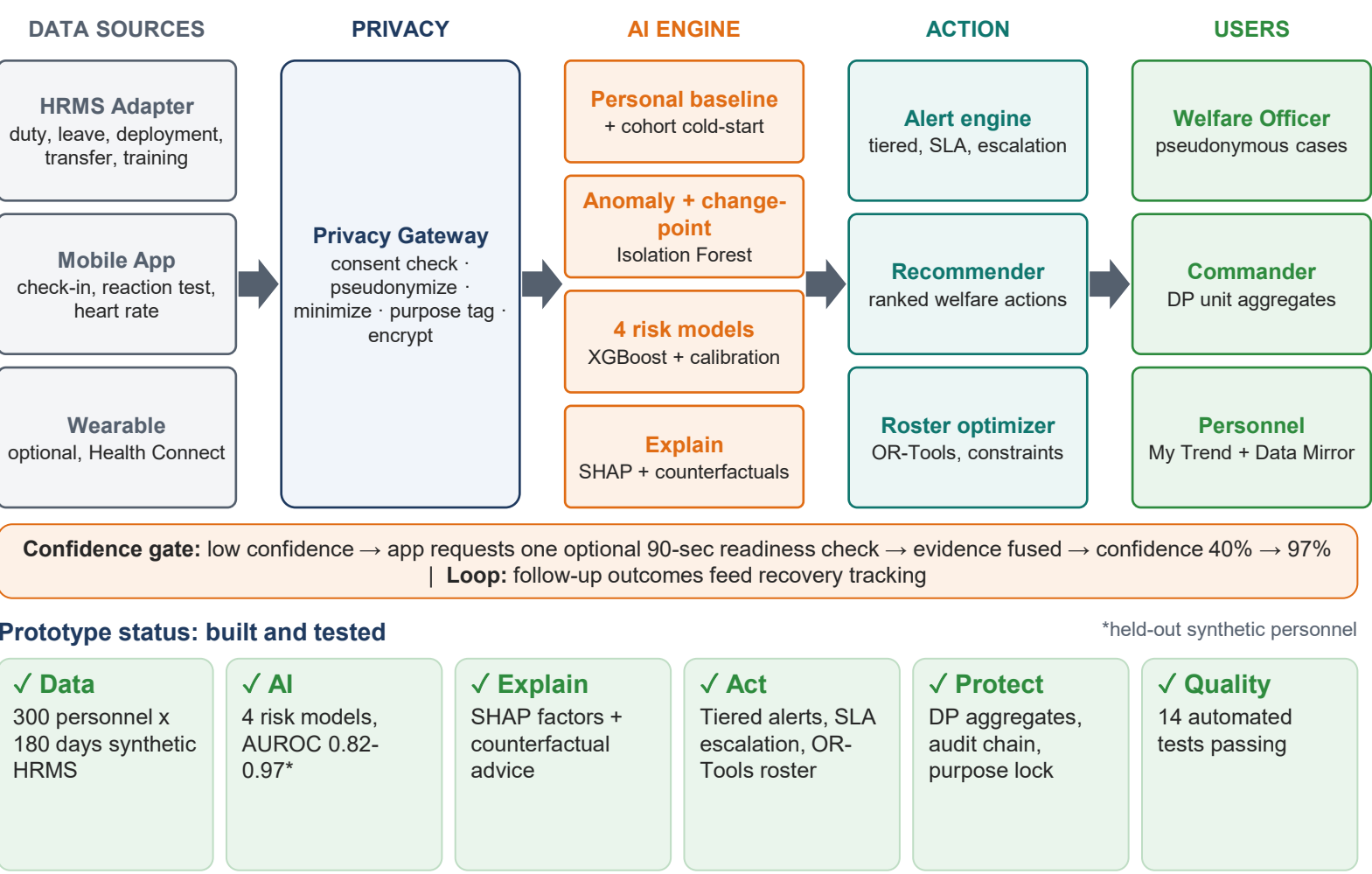
### Working prototype · Welfare Officer Console

Live screen from our running app (React + FastAPI + XGBoost), synthetic data. Case P-104: 4 risk scores, evidence agreement, SHAP factors, counterfactual advice, human decision.

## ❖ Technologies to be used

- Personnel app**  
React 19 PWA, Hindi/English, on-device reaction test; Android via Capacitor
- Dashboards**  
React 19, Vite, Tailwind CSS, Recharts
- Backend API**  
Python FastAPI, Pydantic, JWT (OIDC-ready), OpenAPI
- Data**  
PostgreSQL (production) · SQLite (prototype) via SQLAlchemy
- AI / ML**  
XGBoost x4 + isotonic calibration, SHAP, Isolation Forest, ruptures
- Optimization**  
Google OR-Tools CP-SAT fair-roster solver
- Security**  
Role + unit + purpose checks, SHA-256 hash-chained audit, differential privacy
- Deployment**  
Force data centre or MeghRaj (Gol cloud); open-source, CPU-only, no public LLM

## ❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)



## ❖ Analysis of the feasibility of the idea



### Technical ✓

Working prototype runs end to end today. CPU-only open-source ML; AUROC 0.82-0.97 on held-out synthetic personnel.



### Operational ✓

10-sec daily check-in. Tiered alerts keep load workable (~7% High in a normal unit). Kiosk mode for remote posts.



### Economic ✓

No licence fees; runs on existing force data centres; one platform reusable across all CAPFs.



### Legal & ethical ✓

DPDP Act 2023 aligned consent; non-diagnostic; human-in-the-loop; welfare-only purpose lock.

## ❖ Potential challenges and risks

### False alarms & stigma



Confidence gate + multi-signal agreement + human review; supportive, non-diagnostic language

### No real labelled data



Synthetic generator with hidden strain state; governed pilot with force psychologists (DIPR)

### Low adoption & trust



10-sec check-in, personal benefit first, My Data Mirror, anonymous support option

### Data breach or misuse



On-prem, AES-256, RBAC + ABAC, purpose lock, differential privacy, tamper-evident audit

### Remote posts / no phone



Offline sync, shared kiosk with PIN, HRMS-only mode flagged as low confidence

### Welfare vs mission needs



Optimizer keeps coverage, skill & rest rules; advisory only, commander approves

### What SAHARA will never do

✗ Read chats, calls or social media   ✗ GPS / camera surveillance   ✗ Face or emotion recognition   ✗ Diagnosis or disciplinary use  
 ✗ Public cloud or LLM for personnel data

## ❖ Potential impact on the target audience



### Personnel & families

Early, confidential support; fairer duty spread; a voice without fear of stigma



### Welfare Officers & Counsellors

Prioritized, explainable queue; guided actions; follow-ups tracked automatically



### Unit Commanders

Fatigue hotspots 14 days ahead; balanced rosters; readiness without personal data



### MoHA & Force HQ

Evidence-based welfare planning, counsellor allocation and retention insight

## ❖ Benefits of the solution (social, economic, environmental, etc.)



### Social

*Preventive care, not reactive*

- Fewer stress-related incidents
- Stigma-free help-seeking
- Family welfare needs resolved faster



### Economic

*Protects trained manpower*

- Better retention, lower attrition cost
- Fewer absence & medical days
- Zero licence cost, existing servers



### Operational

*Stronger force readiness*

- Fatigue-aware, fair rosters
- Balanced night duty & deployments
- Data-driven welfare planning



### Environmental & scale

*Green, sovereign, reusable*

- Paperless welfare workflows
- No new hardware; on-prem Indian stack
- Scales to State Police, NDRF/SDRF, Armed Forces



**Live prototype: personnel app in Hindi**  
10-sec check-in · reaction test · help button

## ❖ Details / Links of the reference and research work

### Problem evidence (Government of India)

[1] MHA, Rajya Sabha Unstarred Q.1036 (4 Dec 2024): Attrition and suicides amongst CAPFs. mha.gov.in

### Validated instruments

[2] Topp CW et al. The WHO-5 Well-Being Index: a systematic review. Psychother Psychosom. 2015;84:167–176.

[3] Kristensen TS et al. The Copenhagen Burnout Inventory. Work & Stress. 2005;19(3):192–207.

[4] Basner M, Mollicone D, Dinges DF. Validity of a brief psychomotor vigilance test (PVT-B). Acta Astronautica. 2011;69:949–959.

### Stress signals & occupational context

[5] Kim HG et al. Stress and heart rate variability: a meta-analysis. Psychiatry Investig. 2018;15(3):235–245.

[6] Coppetti T et al. Accuracy of smartphone apps for heart rate measurement. Eur J Prev Cardiol. 2017;24(12):1287–1293.

[7] Violanti JM et al. Police stressors and health: a state-of-the-art review. Policing. 2017;40(4):642–656.

### AI / ML & privacy methods

[8] Chen T, Guestrin C. XGBoost: a scalable tree boosting system. KDD 2016. | Liu FT et al. Isolation Forest. IEEE ICDM 2008.

[9] Lundberg SM, Lee SI. A unified approach to interpreting model predictions (SHAP). NeurIPS 2017.

[10] Wachter S et al. Counterfactual explanations without opening the black box. Harvard J Law & Tech. 2018.

[11] Dwork C, Roth A. The Algorithmic Foundations of Differential Privacy. 2014.

### Policy & platforms (India)

[12] Digital Personal Data Protection Act, 2023: meity.gov.in

[13] Tele-MANAS national tele-mental health helpline 14416: telemanas.mohfw.gov.in

[14] Google OR-Tools CP-SAT solver: developers.google.com/optimization

## Gap analysis from our research

### MHA's current measures [1] → what SAHARA adds

- **Transparent leave policy** → leave-deprivation tracking and alerts
- **Regulating duty hours** → fair-roster optimizer with rest rules
- **Officer–troop grievance interaction** → Welfare Need module with SLA
- **Stress talks, yoga, counselling** → early, targeted referral to who needs it

### Typical tech approach → SAHARA

- **Needs wearables** → works on any phone + existing HRMS data
- **One 'stress score'** → 4 calibrated risks, each with confidence
- **Chat / face monitoring** → no surveillance; consent per data type
- **Alert only** → counterfactual advice + fair roster + recovery loop
- **Claimed accuracy** → measured on held-out data, published honestly